Distance and Scale Interface under Observation Maps

Global pseudometrics, quotient geometry, matched resolutions, and calibrated shells

Stassis Research Program

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Status. This document replaces *Unit Distances under Observation Maps v0.1* as the distance-scale reference module for GO Core v0.2. A numerical value "1" is never compared with a physical distance without a declared reference radius. The global observed pseudometric is separated from the local pullback tensor, all resolution maps are typed, and exact-shell statements are separated from tolerant or noisy decisions.

Abstract

Let $O : (X, d_X) \to (Y, d_Y)$ be an observation map between metric spaces. The globally observed distance on hidden states is the pseudometric $d_O(x, x') = d_Y(Ox, Ox')$. Its zero classes are the observation fibers, and the metric quotient is canonically isometric to the image $O(X)$. This global construction is distinct from the local pullback quadratic form O^*g_Y; the corresponding path pseudometric can exceed the ambient image distance unless a minimizing image geodesic is liftable. A distance shell is defined relative to a positive, typed radius r_*, while finite resolution and measurement error require a tolerance τ and an explicit decision rule. Under coherent changes of unit, d_Y, r_*, τ, and the output resolution rescale together, leaving all normalized predicates invariant. Lipschitz observations propagate input resolution to output resolution and compose multiplicatively. The orthogonal-projection cylinder, fiber multiplicities in observed unit-distance graphs, and periodic half-step aliasing are derived as applications. The module is an interface built from standard metric and differential geometry; it does not solve the classical Erdos unit-distance problem.

Contents

1 Scope, objects, and claim boundary

The purpose of this module is to make every use of "distance" and "scale" in Geometry of Observation type-checkable. It supplies a common interface for metric entropy, ruler traces, mechanics, and later observation modules. It does not impose a physical ontology on the hidden space.

Boundary 1.1 (Claim boundary). An observation-induced pseudometric is a standard pullback construction. Fiber multiplicity can create quadratically many hidden-state pairs, but this is not a new lower bound for the classical planar unit-distance function. A degenerate pullback tensor does not by itself imply a physical extra dimension.

Definition 1.2 (Typed metric datum)**.** A typed metric datum is a tuple

$$\mathfrak{M}_X = (X, d_X, U_X),$$

where $d_X : X \times X \to \mathbb{R}_{\geq 0} U_X$ is a metric and U_X names its scale type. For a physical spatial metric, U_X may be metres; for a normalized abstract metric, $U_X = 1$.

The notation $\mathbb{R}_{\geq 0} U_X$ is a bookkeeping device: it records that d_X is length-valued (or more generally scale-valued), not an untyped real number. Ratios of two positive quantities with the same scale type are dimensionless.

Definition 1.3 (System and protocol passports)**.** The intrinsic descriptor and the observation protocol are distinct:

$$\xi_X = (X, d_X, \operatorname{diam} X, \text{intrinsic regularity data}, \ldots),$$

$$\lambda = (O, d_Y, r_*, \varepsilon_X, \varepsilon_Y, \tau, Q, \mathsf{E}, \mathsf{U}, \ldots).$$

Here $r_* > 0$ is a declared output-space reference radius, $\varepsilon_X, \varepsilon_Y > 0$ are input and output resolutions, $\tau \geq 0$ is a decision tolerance, Q is a quantizer or partition, E is an estimator or direct readout, and U is an uncertainty model.

An intrinsic diameter is not an instrument resolution. A shell radius is not a regression reference length. The same physical dimension therefore does not imply the same semantic role.

2 The global observed pseudometric

Definition 2.1 (Observed pseudometric)**.** Let (Y, d_Y) be a metric space and let $O : X \to Y$ be any map. Define

$$d_O(x, x') := d_Y\big(O(x), O(x')\big).$$

Theorem 2.2 (Pseudometric and separation criterion)**.** *The function d_O is a pseudometric on X. It is a metric if and only if O is injective.*

Proof. Nonnegativity and symmetry follow from d_Y. For $x, x', x'' \in X$,

$$d_O(x, x'') = d_Y(Ox, Ox'') \leq d_Y(Ox, Ox') + d_Y(Ox', Ox'') = d_O(x, x') + d_O(x', x'').$$

Thus the triangle inequality holds. Moreover, $d_O(x, x') = 0$ if and only if $O(x) = O(x')$. Hence separation is equivalent to injectivity. □

Definition 2.3 (Observation equivalence)**.** Write

$$x \sim_O x' \iff O(x) = O(x').$$

Let $X/{\sim_O}$ denote the set of observation fibers.

Theorem 2.4 (Canonical quotient metric)**.** *The formula*

$$\bar{d}_O([x], [x']) := d_Y(Ox, Ox')$$

is well-defined and is a metric on $X/{\sim_O}$. The map

$$\iota_O : X/{\sim_O} \longrightarrow O(X), \qquad [x] \longmapsto O(x),$$

is an isometry onto the metric subspace $O(X) \subseteq Y$.

Proof. Changing representatives leaves both images unchanged, so the formula is well-defined. The pseudometric axioms descend from theorem 2.2; zero distance now implies equality of classes. The map ι_O is a bijection and preserves the displayed distance. □

$$\begin{tikzcd}
\begin{array}{c} X \\ \text{hidden states} \end{array} \arrow[r, "\text{quotient}"] \arrow[rr, bend right, "O"'] & \begin{array}{c} X/{\sim_O} \\ \text{observable classes} \end{array} \arrow[r, "\iota_O\ \text{isometry}"] & \begin{array}{c} O(X) \subseteq Y \\ \text{metric image} \end{array}
\end{tikzcd}$$

3 Local pullback form versus global distance

Suppose now that X and Y are smooth manifolds, Y has Riemannian metric g_Y, and $O : X \to Y$ is smooth.

Definition 3.1 (Pullback quadratic form)**.**

$$g_O := O^* g_Y, \qquad (g_O)_x(v, w) = (g_Y)_{O(x)}\big(dO_x v, dO_x w\big).$$

Proposition 3.2 (Degeneracy directions)**.** *g_O is positive semidefinite and*

$$\ker dO_x \subseteq \ker (g_O)_x.$$

If g_Y is positive definite, equality holds.

Proof. Positive semidefiniteness follows from $g_O(v, v) = g_Y(dO\, v, dO\, v) \geq 0$. If $dO\, v = 0$, then $g_O(v, w) = 0$ for every w. Conversely, if $g_O(v, v) = 0$, positive definiteness of g_Y gives $dO\, v = 0$. □

For every piecewise C^1 curve γ in X,

$$L_{g_O}(\gamma) = L_{g_Y}(O \circ \gamma).$$

Define the extended path pseudometric

$$\rho_O(x, x') := \inf_{\gamma : x \rightsquigarrow x'} L_{g_O}(\gamma),$$

with value $+\infty$ if there is no admissible path.

Proposition 3.3 (Ambient versus intrinsic image distance)**.** *For all path-connected pairs,*

$$d_O(x, x') \le \rho_O(x, x').$$

Equality holds if a d_Y-minimizing geodesic from $O(x)$ to $O(x')$ lies in $O(X)$ and has a lift joining x to x'.

Proof. Every image curve joining $O(x)$ to $O(x')$ has length at least their ambient Riemannian distance. Taking the infimum gives the inequality. A lift of a minimizing geodesic realizes equality. □

Remark 3.4. The local tensor g_O, the path pseudometric ρ_O, and the global pullback pseudometric $d_O = d_Y \circ (O \times O)$ are therefore not interchangeable without the hypotheses of proposition 3.3. This corrects the ambiguity in v0.1.

4 Reference shells and calibrated decisions

Definition 4.1 (Exact and tolerant shells)**.** Let $r_* > 0$ and $\tau \ge 0$ have the same scale type as d_Y. For $x \in X$, define

$$S_O(x; r_*) := \left\{ x' \in X : d_O(x, x') = r_* \right\},$$

$$S_O^\tau(x; r_*) := \left\{ x' \in X : \left| d_O(x, x') - r_* \right| \le \tau \right\}.$$

The exact shell is the case $\tau = 0$.

Definition 4.2 (Observed distance graph)**.** For a finite set $A = \{x_1, \ldots, x_n\} \subseteq X$, define

$$G_{O,r_*,\tau}(A) := \big(A, E_{O,r_*,\tau}\big),$$

$$\{x_i, x_j\} \in E_{O,r_*,\tau} \quad \Longleftrightarrow \quad |d_O(x_i, x_j) - r_*| \le \tau.$$

Boundary 4.3 (Exact equality and measured data). For continuously noisy data, exact numerical equality is generally not a stable event. The mathematical shell $\tau = 0$ remains valid, but an empirical classifier must declare a tolerance or a hypothesis test.

Proposition 4.4 (Deterministic error guarantee)**.** *Suppose an estimated distance $\widehat{d}_{ij}$ obeys*

$$\left| \widehat{d}_{ij} - d_O(x_i, x_j) \right| \le u$$

with a declared bound $u \ge 0$. The rule

$$\left| \widehat{d}_{ij} - r_* \right| \le \tau$$

has the following guarantees:

(i) *every exact-shell pair is accepted if* $\tau \geq u$*;*

(ii) *every accepted pair satisfies* $|d_O(x_i, x_j) - r_*| \leq \tau + u$*.*

Proof. Both statements are direct applications of the triangle inequality. □

If u is probabilistic rather than deterministic, the confidence level, dependence structure, and multiple-testing policy belong to the protocol.

5 Coherent unit covariance

Let $a > 0$ be a change of numerical length unit. Define

$$d'_Y = a\, d_Y, \qquad r'_* = a\, r_*, \qquad \tau' = a\, \tau, \qquad \varepsilon'_Y = a\, \varepsilon_Y.$$

Theorem 5.1 (Unit covariance of shells and graphs)**.**

$$S^{\tau'}_{O,d'_Y}(x; r'_*) = S^{\tau}_{O,d_Y}(x; r_*),$$

and therefore

$$G_{O,r'_*,\tau'}(A; d'_Y) = G_{O,r_*,\tau}(A; d_Y).$$

Proof.

$$\left|a\, d_O(x, x') - a\, r_*\right| \leq a\, \tau \quad \Longleftrightarrow \quad \left|d_O(x, x') - r_*\right| \leq \tau.$$

□

The normalized quantities

$$\bar{d}_O = \frac{d_O}{r_*}, \qquad \bar{\tau} = \frac{\tau}{r_*}, \qquad \bar{\varepsilon}_Y = \frac{\varepsilon_Y}{r_*}$$

are invariant under the same coherent rescaling. A formula $d_O(x, x') = 1$ is admissible only after the convention $r_* = 1\, U_Y$ has been declared. The typed formula is $d_O(x, x') = r_*$.

Definition 5.2 (Dimensionless log-scale coordinate)**.** For a positive reference scale ℓ_* and resolution $\varepsilon > 0$,

$$s_b(\varepsilon; \ell_*) := \log_b \frac{\ell_*}{\varepsilon}, \qquad b > 1.$$

No logarithm of a bare physical length is admissible.

6 Matched resolutions under Lipschitz observation

Definition 6.1 (Typed Lipschitz converter)**.** An observation $O : (X, d_X) \to (Y, d_Y)$ is L_O-Lipschitz if

$$d_Y(Ox, Ox') \leq L_O\, d_X(x, x')$$

for all x, x'. Semantically,

$$[L_O] = U_Y / U_X.$$

Proposition 6.2 (Resolution propagation)**.** *If an input cell has* d_X*-diameter at most* ε_X*, its image has* d_Y*-diameter at most* $L_O \varepsilon_X$*. Hence a conservative matched output resolution satisfies*

$$L_O \varepsilon_X \leq \varepsilon_Y.$$

Proof. Apply the Lipschitz inequality to every pair in the input cell. □

Theorem 6.3 (Composition law). *If $O_1 : (X, d_X) \to (Y, d_Y)$ is L_1-Lipschitz and $O_2 : (Y, d_Y) \to (Z, d_Z)$ is L_2-Lipschitz, then $O_2 \circ O_1$ is L_2L_1-Lipschitz. Matched scales propagate as*

$$\varepsilon_Y \geq L_1\varepsilon_X, \qquad \varepsilon_Z \geq L_2\varepsilon_Y \quad \Longrightarrow \quad \varepsilon_Z \geq L_2L_1\varepsilon_X.$$

Proof. Substitute the first Lipschitz inequality into the second. The scale statement follows by transitivity. □

This is the distance-scale bridge to the matched covering defect in the Metric Entropy v0.2 module.

7 Orthogonal projection: cylinder and quotient

Let

$$\pi : \mathbb{R}^3 \to \mathbb{R}^2, \qquad \pi(x, y, z) = (x, y),$$

with Euclidean metrics.

Proposition 7.1 (Observed shell under projection). *For $X_0 = (x_0, y_0, z_0)$ and $r_* > 0$,*

$$S_\pi(X_0; r_*) = \left\{(x, y, z) : (x - x_0)^2 + (y - y_0)^2 = r_*^2\right\}.$$

Thus the exact observed shell is a circular cylinder of radius r_*.

Proof. The condition $d_\pi(X, X_0) = \|\pi(X) - \pi(X_0)\|_2 = r_*$ is precisely the displayed equation; z is free. □

Corollary 7.2 (Hidden distance profile). *If $d_\pi(X, X') = r_*$, then*

$$\|X - X'\|_2^2 = r_*^2 + (z - z')^2.$$

One observed shell radius therefore corresponds to a continuum of hidden Euclidean distances greater than or equal to r_*.

The quotient $\mathbb{R}^3/{\sim_\pi}$ is isometric to $\mathbb{R}^2$; the cylinder is the preimage of a planar circle, not a new type of planar unit-distance configuration.

8 Fiber multiplicity and counting rules

Two non-equivalent counts must be declared:

(a) *visible count*: replace A by the set of distinct images $O(A)$ and count image pairs;

(b) *hidden-multiplicity count*: count pairs of distinct hidden states whose images satisfy the shell predicate.

Theorem 8.1 (Complete bipartite fiber lift). *Let $F_a = O^{-1}(a)$ and $F_b = O^{-1}(b)$, with $d_Y(a, b) = r_*$. If a finite sample contains m states in F_a and k states in F_b, its hidden-multiplicity graph contains $K_{m,k}$ and at least mk shell edges.*

Proof. Every cross-fiber pair has observed distance $d_Y(a, b) = r_*$. □

Proposition 8.2 (Reduction to the visible problem). *If O is injective on a finite sample A and the count is taken over distinct images, then*

$$|E_{O,r_*,0}(A)| = \#\left\{\{a, b\} \subset O(A) : d_Y(a, b) = r_*\right\}.$$

Proof. The restriction $O|_A$ is a vertex bijection preserving the edge predicate. □

9 Dimensionally valid periodic and half-step readout

Let a hidden diameter ladder be

$$d_n = d_0 + n\Delta d, \qquad n \in \mathbb{Z},$$

where d_0 and Δd are lengths. Let $P > 0$ be a physical period and ϕ_0 an angle. Define

$$h(n) = \cos\left(2\pi \frac{d_n - d_0}{P} + \phi_0\right).$$

The trigonometric argument is an angle because $(d_n - d_0)/P$ is dimensionless.

Theorem 9.1 (Rational-step factorization)**.** *If*

$$\frac{\Delta d}{P} = \frac{p}{q}, \qquad p \in \mathbb{Z},\ q \in \mathbb{N},\ \gcd(p,q) = 1,$$

then h factors through $\mathbb{Z}/q\mathbb{Z}$.

Proof. For every n,

$$h(n+q) = \cos\left(2\pi\frac{p(n+q)}{q} + \phi_0\right) = \cos\left(2\pi\frac{pn}{q} + 2\pi p + \phi_0\right) = h(n).$$

□

The theorem states factorization, not injectivity on the q residue classes; cosine symmetry may identify additional classes.

Corollary 9.2 (Half-step parity)**.** *If $\Delta d = P/2$ and $\phi_0 = 0$, then*

$$h(n) = \cos(\pi n) = (-1)^n,$$

so the readout is exactly the parity quotient $\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$.

This replaces the dimensionally invalid expression $d_n = d_0 + n/2$ unless a length unit for the half-step has first been declared.

10 Unit-distance graphs as an application

For $P \subset \mathbb{R}^2$ finite and $r_* > 0$, write

$$U_{r_*}(P) := \#\left\{\{p,q\} \subset P : \|p-q\|_2 = r_*\right\}.$$

Euclidean similarity $p \mapsto p/r_*$ gives

$$\max_{|P|=n} U_{r_*}(P) = \max_{|P|=n} U_1(P).$$

Thus the classical extremal function is independent of the chosen positive reference radius after coherent rescaling.

For an observation map $O : X \to \mathbb{R}^2$, define instead

$$U^{\text{hidden}}_{O,r_*}(A) := \# E_{O,r_*,0}(A).$$

This quantity can be quadratic because of theorem 8.1. It is a different problem: multiplicity inside observation fibers is part of the data.

Boundary 10.1 (Erdos firewall). The present interface proves no new asymptotic upper or lower bound for the classical planar unit-distance function. It only states the data needed to define observed-distance variants without dimensional or counting ambiguity.

11 Normative protocol ledger

For a static theorem with exact metric data, the stochastic-channel field is recorded as `explicit-noiseless`, while temporal resolution and observation horizon are recorded as `not-applicable`. They become active fields for time-dependent geometry. This module defines no scalar defect, so `defect-normalizations` is recorded as `not-applicable`; a downstream distortion or decision loss must provide its own typed normalization.

Field	Type	Required declaration
Hidden space	set, manifold, or state space	admissible points and equivalence before observation
Hidden metric d_X	U_X	metric convention and domain
Observation O	map $X \to Y$	deterministic, stochastic, or composed channel role
Noise/channel policy	kernel or exact flag	stochastic law or explicit noiseless statement
Observed metric d_Y	U_Y	ambient or intrinsic convention
Observed pseudometric d_O	U_Y	global formula $d_Y(Ox, Ox')$
Reference radius r_*	U_Y, positive	shell or incidence target
Input resolution ε_X	U_X, positive	quantizer or cover convention
Output resolution ε_Y	U_Y, positive	matched-scale rule
Temporal resolution	time or not applicable	required for time-dependent geometry
Observation horizon	time or not applicable	required for a dynamic trace
Tolerance τ	U_Y, nonnegative	exact, bounded-error, or hypothesis-test policy
Uncertainty u or law	U_Y or distribution	confidence/dependence statement
Counting rule	discrete	visible images or hidden multiplicities
Reference scale ℓ_*	same type as scale variable	normalization and zero policy
Logarithm base b	dimensionless, $b > 1$	every logarithmic trace
Estimator	declared map	output, window, and uncertainty summary
Defect normalization	typed ratio or not applicable	required before any scalar loss aggregation

12 Claim audit

Statement	Status	Necessary hypotheses
$d_O = d_Y \circ (O \times O)$ is a pseudometric	theorem	d_Y metric, O a map
$X/\sim_O$ is isometric to $O(X)$	theorem	quotient by equal observations
$g_O = O^* g_Y$ determines the same global distance as d_O	not generally true	requires a suitable lift of minimizing image paths
Observed exact shells are unit-covariant	theorem	coherent rescaling of metric, radius, and tolerance
Fiber multiplicity produces $K_{m,k}$	theorem	two finite sampled fibers at the target separation
Half-step readout recovers the hidden state	false	parity readout is many-to-one
Observation-map graphs solve the classical unit-distance problem	not claimed	the classical problem is only the injective visible special case

13 Conclusion

The common distance-scale object is not a bare number. It is

$$\big(X, d_X, O, Y, d_Y, d_O, r_*, \varepsilon_X, \varepsilon_Y, \tau, \mathsf{U}, \text{counting rule}\big).$$

The global pseudometric identifies exactly the observation fibers; its quotient is the metric image. Physical comparisons use typed radii and tolerances, logarithms use reference ratios, and resolutions propagate through declared Lipschitz converters. These rules are the dependency needed by scale-trace modules such as Mandelbrot rulers.

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